

From Half-Log Rank Clocks to Schur Packet Bootstrap

A Ten-Layer Review and the Revised Stage-A Frontier

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Abstract

RH-82–RH-90 pursued the preferred postblock effective-rank corridor from an endpoint rank clock to a full-reference-free Schur certificate. This tenth paper extracts the resulting all-level sufficient theorem, audits the nine dependencies, and revises the Stage-A/A5 frontier.

For normalized memory Gramians $G_j = X_j^* X_j / \|X_j\|_2^2 + \eta G_{j-1}$, suppose a nondecreasing clock-rank law and the RH-90 Schur condition give $E_{j+1} \leq \rho E_j$ for m consecutive late updates, where E_j is the optimal packet tail and $0 < \rho < 1$. The Schur-to-effective-rank bootstrap theorem gives

$$E_{b+m} \leq \frac{\rho^m}{1 - \eta}, \quad \frac{\tau_{r_{b+m}}(X_{b+m})}{\|X_{b+m}\|_2} \leq \frac{\rho^{m/2}}{\sqrt{1 - \eta}}.$$

At the audited values $\eta = 1/512$ and $\rho = 0.24$, 20 updates suffice for a 10^{-6} relative tail and 39 for 10^{-12} .

The nine archived summaries contain 29 theorem flags, all green in their stated scopes. The positive chain is: half-log clock, captured energy, strict-prefix packet, normalized memory, scalar injection, predictor-corrector factorization, rank-one Ritz enrichment, and Schur sign. The branch-specific negative results rule out coordinate identity, unweighted prefix Gramians, tail-angle perturbation, global-norm contraction, and uniform point-packet contraction as shortcuts.

The revised completion frontier is not a claim that Stage A is closed. Its two alternatives are the RH-75 full-block law or a three-part Schur packet bundle: uniform late Schur contraction, polylogarithmic reduced packet future, and the inherited finite-prefix/normalization/observability bridge. The A5 moving-cloud projection, coefficient bridge, and uniform complement remain unchanged. No Hilbert–Polya or Riemann Hypothesis conclusion is made.

Keywords: effective rank; Schur certificate; dynamic packet; dependency frontier; validated numerics; small noise.

MSC 2020: 47A75; 47B10; 37C30; 65G20.

1 How the route changed

RH-82 sharpened the endpoint resolution theorem to a half-logarithmic excess-rank tail [2]. RH-83 characterized the optimal singular factorization while rejecting coordinate identity [1]. RH-84 then showed that the downstream problem needs only captured tail energy, not factorization or termwise singular control [4].

RH-85 constructed a strict-prefix dynamic packet and proved that unweighted prefix energy can fail [10]. RH-86 introduced a trace-normalized memory packet and found that principal-angle bounds are quantitatively unusable [5]. RH-87 reduced packet drift to a scalar Rayleigh injection recursion [3].

RH-88 separated old-packet prediction from variational correction and closed the global-norm and uniform point-packet shortcuts at the anchors [6]. RH-89 found that one complement direction plus an $(r + 1)$ -dimensional Ritz solve recovers almost all reference correction [7]. RH-90 replaced that full reference by a directly validated small Schur sign [9].

2 Schur-to-effective-rank bootstrap theorem

Let

$$\hat{G}_j = \frac{X_j^* X_j}{\|X_j\|_2^2}, \quad G_j = \hat{G}_j + \eta G_{j-1}, \quad G_{-1} = 0,$$

with $0 \leq \eta < 1$. Let $E_{j,r}$ be the optimal rank- r tail of G_j .

Theorem 2.1 (Schur-to-effective-rank bootstrap theorem). *Let r_j be nondecreasing. Suppose that for $j = b, \dots, b + m - 1$, either the old packet already meets the target or an RH-90 Schur trial certifies*

$$E_{j+1, r_{j+1}} \leq \rho E_{j, r_j}, \quad 0 < \rho < 1. \quad (1)$$

Then

$$E_{b+m, r_{b+m}} \leq \frac{\rho^m}{1 - \eta}, \quad (2)$$

$$\frac{\tau_{r_{b+m}}(X_{b+m})}{\|X_{b+m}\|_2} \leq \frac{\rho^{m/2}}{\sqrt{1 - \eta}}. \quad (3)$$

Consequently a tolerance ε is reached once

$$m \geq \frac{2 \log(1/(\varepsilon \sqrt{1 - \eta}))}{|\log \rho|}.$$

Proof. Every $\hat{G}_j$ has trace one, hence $\text{tr } G_b \leq \sum_{k=0}^b \eta^k \leq (1 - \eta)^{-1}$ and $E_{b, r_b} \leq (1 - \eta)^{-1}$. Iterating (1) gives (2). The current snapshot is a weight-one summand of the memory objective, so its optimal relative squared residual is at most $E_{b+m, r_{b+m}}$. Taking square roots proves (3) and the update count. $\square$

Corollary 2.2 (Conditional RH-78 residual gate). *Assume the clock rank is $O(\log(1/\sigma))$, the Schur updates in Theorem 2.1 persist for the required late window, and $\sqrt{G_\sigma} \|X_\sigma\|_2$ is polylogarithmic. Then the observability-transferred effective-rank residual is polylogarithmic, and may be made power-small if the late window is $\Omega(\log(1/\sigma))$.*

This corollary supplies only the residual term in the RH-78 effective-rank corridor [11]. The reduced packet future and finite prefix must still be proved polylogarithmic.

3 Nine-paper ledger and negative markers

The five branch-specific negative results are durable route markers, not global no-go theorems. In particular, the failure of tail-angle continuity does not conflict with stable captured energy, and the failure of old-packet contraction does not conflict with corrected packet contraction.

4 Validated bootstrap budget

The review hashes all nine summaries and evaluates the bootstrap at 256-bit precision with $\eta = 1/512$ and $\rho = 0.24$.

The count is logarithmic in inverse tolerance. It does not show that the Schur sign persists for 39 physical updates; that persistence is precisely the new all-level theorem wall.

Paper	Durable object	Route effect	Remaining boundary
82	half-log tail	rank clock	physical transfer
83	optimal factorization	strength audit	coordinate no-go
84	Ky Fan tail	weaker invariant	all-level energy
85	prefix packet	dynamic construction	unweighted no-go
86	normalized memory	gap-free online packet	angle no-go
87	injection recursion	scalar reduction	injection law
88	predictor/corrector	correction dividend	norm/point no-go
89	one-direction Ritz	small corrector	cross enrichment
90	Schur trial	reference-free certificate	uniform sign

Table 1: RH-82–RH-90. Negative verdicts apply only to the named shortcut.

relative tolerance	minimal updates	certified bound
10^{-2}	7	6.779×10^{-3}
10^{-4}	13	9.372×10^{-5}
10^{-6}	20	6.347×10^{-7}
10^{-8}	26	8.774×10^{-9}
10^{-10}	33	5.942×10^{-11}
10^{-12}	39	8.214×10^{-13}

Table 2: Minimality is certified by checking that the preceding update remains above the requested tolerance.

5 Revised completion frontier

Let L be the all-level full-block law, S the uniform late Schur update law, R the polylogarithmic reduced packet future, and O the inherited finite-prefix/normalization/observability bridge. With the finite chain closed in its archived scope, the revised sufficient Stage-A formula is

$$A_1 = L \vee (S \wedge R \wedge O).$$

Its minimal open bundles are $\{L\}$ and $\{S, R, O\}$. This is the revised completion frontier: RH-82–90 greatly sharpen S , but do not prove S , R , or O uniformly.

For relative A5, append the unchanged moving-cloud gates P, C, U from RH-81 [8]. The two sufficient bundles become

$$\{L, P, C, U\}, \quad \{S, R, O, P, C, U\}.$$

No canonical scattering completion, self-adjoint generator, intrinsic $T \log T$ law, prime-power trace formula, completed-zeta identity, Hilbert–Polya operator, zeta-zero identification, or Riemann Hypothesis result is established here.

References

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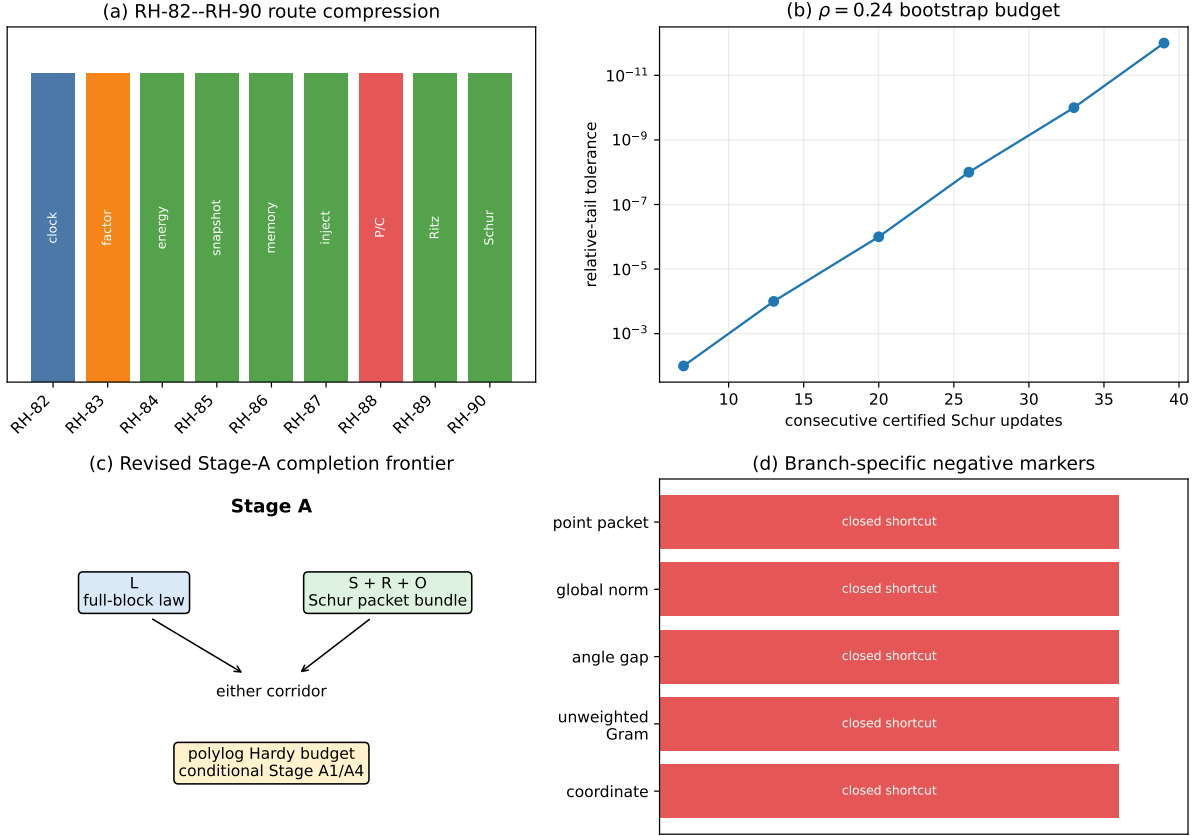


Figure 1: The nine-layer compression, contraction budget, revised Stage-A frontier, and branch-specific negative markers.

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